

LAB #14: COMPLEX COMPUTING ACTIVITY (CCA)

Project Title: HealthGuard AI: Integrated Medical Triage & Clinical Decision Support System

Domain: Healthcare Analytics / Predictive Medicine

Student Name: Santosh Kumar

Date: 13/1/26

1. ABSTRACT

In modern healthcare, rapid patient prioritization (Triage) is critical for saving lives. This project presents **HealthGuard AI**, an end-to-end Machine Learning application designed to classify patient risk levels (Low, Medium, and High) based on real-time physiological data. The system addresses common real-world challenges such as data imbalance and the "black-box" nature of AI by integrating data engineering, ensemble modeling, and model interpretability.

2. PROBLEM DESCRIPTION & DOMAIN

Medical Triage involves the assessment of patients to determine the urgency of their need for care. Manual triage can be prone to human error and fatigue. This project automates the process using features like **Systolic Blood Pressure, Oxygen Saturation (SpO2), Heart Rate, and Body Temperature**. The goal is to provide a reliable, second-opinion tool for clinicians that is transparent and data-driven.

3. INTEGRATION OF LAB CONCEPTS

To demonstrate mastery of the curriculum, this project integrates three distinct concepts from the Lab Manual:

Concept Integrated	Lab Ref.	Implementation Detail
Data Preprocessing	Lab 01	Handled clinical feature scaling using StandardScaler and addressed the "High-Risk" minority class using SMOTE (Synthetic Minority Over-sampling).
Random Forest Classifier	Lab 05	Utilized as the core predictive engine. Random Forest was chosen over simpler models due to its ability to handle non-linear clinical relationships.

Concept Integrated	Lab Ref.	Implementation Detail
K-Means Clustering	Lab 10	Implemented as an unsupervised anomaly detection layer to group similar patient profiles and flag "Outlier" health readings before classification.

4. SYSTEM ARCHITECTURE & PIPELINE

The system follows a 4-stage pipeline:

1. **Data Generation Layer:** A custom script simulates Electronic Medical Records (EMR) for 1,500 patients.
2. **Engineering Layer:** Handles missing values, performs feature scaling, and balances classes to ensure the model isn't biased toward "Low Risk" cases.
3. **Inference Layer:** An ensemble of Decision Trees (Random Forest) processes the vitals to predict a Triage category.
4. **Explainability Layer:** Generates feature importance plots to show the doctor which vital sign (e.g., low Oxygen) was the primary reason for a "High Risk" alert.

5. EXPERIMENTAL RESULTS

- **Accuracy:** The model achieved an overall accuracy of **~92%** on the balanced test set.
- **Precision/Recall:** High recall was prioritized for the "High Risk" class (Class 2) to ensure no critical patients are missed.
- **Key Findings:** Feature importance analysis revealed that **Oxygen Saturation** and **Blood Pressure** were the most significant predictors of patient deterioration.

6. CHALLENGES & LIMITATIONS

- **Data Realism:** While the synthetic data mimics medical trends, real clinical data contains more noise and missing entries.
- **Latency:** Real-time pipelines require optimization to handle thousands of concurrent patient streams.
- **Ethics:** We excluded demographic data (Gender/Race) to prevent the model from learning societal biases, ensuring fair treatment for all patients.

7. SETUP & RUN INSTRUCTIONS

1. **Environment:** Requires Python 3.8+ and libraries: pandas, scikit-learn, imblearn, matplotlib.
2. **Command:** Run `python data_generator.py` to initialize the database.
3. **Command:** Run `python app.py` to train the model and generate the Interpretability Report.

8. CONCLUSION

This Complex Computing Activity successfully integrates the foundational concepts of Machine Learning into a cohesive, high-impact application. By combining unsupervised learning for anomaly detection with supervised ensemble methods, the project demonstrates a robust approach to solving real-world healthcare challenges.